

# CaSiNo Summary

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# Abstract

- Topic: Advance development of practical negotiation systems
- CaSiNo: Corpus containing negotiations between campsite neighbors
- Aims to make a connection between dialogue behaviors and level of success of negotiation
- Propose/evaluate multi-task framework to recognize negotiation strategies
- Found multi-task learning improves the performance for all strategy labels

# Introduction

- Automatic Negotiation provide cost-effective social skill training and assistant
- Negotiation requires understanding the partners motives along w/ effective reasoning and communication
- Prior work in human-machine negotiation often uses predefined menu of options
- While structured and traceable, most dialogue in these settings reduce to an exchange of offers
- Items are arbitrarily defined, no semantic context around the items
- Fails to capture many realistic aspects of negotiation
- Other end, produces rich open-ended dialogue but makes evaluation non-trivial
- Designed Negotiation Task: tractable while resulting in rich dialogues
- Two participants take the role of camp-site neighbors and negotiate for additional Food, Water, and Firewood based on preferences and requirements
- As well as collect satisfaction from the outcome

# CaSiNo Dataset

- Crowd Source from Amazon Mechanical Turk with diverse participant group
- Because the average person is bad at negotiating, a [training video](#) is given to participants
- Requirements: Semantically meaningful, symmetric task, symmetric items
- Task known as the multi-bargaining task
- Diversity is encouraged by broad definition of items
- Inherent biases can influence symmetry, items are “extra”
- Three items are assigned a random priority order permutation, high, med, low
- Two participants are randomly paired, after agreement they submit a deal or walk away
- Evaluation -> sum of points over items {high: 5, med: 4, low: 3}
- Post survey -> How satisfied are you with the negotiation outcome?, How much do you like your opponent?

| Preferences & Arguments                                                      |                                                            |
|------------------------------------------------------------------------------|------------------------------------------------------------|
| P1                                                                           | P2                                                         |
| <i>High:</i> Water: We like to go on runs and it increases the need of this. | <i>High:</i> Food: Food really increases everyones morale. |
| <i>Medium:</i> Food: Food overall is a good mood booster.                    | <i>Medium:</i> Firewood: We like to have a large fire.     |
| <i>Low:</i> Firewood: We do not care for fire and it is not necessary to us. | <i>Low:</i> Water: We don't drink water that often.        |

  

| Conversation                                                                                                                                                        | Annotation                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>P1:</b> How are you today? Did you have any preferences on the supplies we will be trading?                                                                      | Small-Talk, Coordination, Elicit-Pref          |
| <b>P2:</b> I am good. How about yourself? I think I would like some firewood to start off with. We like to have bigger fires. What about you?                       | Small-Talk, Self-Need, Other-Need, Elicit-Pref |
| <b>P1:</b> I am good as well. That is good to hear that you like to have bigger fires as we do not care much for that. We would much rather have some extra water.  | Small-Talk, Empathy, No-Need                   |
| <b>P2:</b> Water is a little important to us too though , if possible maybe we can split that or maybe we can get some more food in replacement.                    | Coordination                                   |
| <b>P1:</b> That may be possible.... 😊 What did you have in mind for the food replacement?                                                                           | Non-strategic                                  |
| <b>P2:</b> You can have all the water if we can have all the food?                                                                                                  | Non-strategic                                  |
| <b>P1:</b> I dont think I am okay with that 😞 . Food is essential to our groups morale when camping. We would like 1 additional food preferably.                    | Self-Need, Other-Need                          |
| <b>P2:</b> Well you guys did say you did not care much about large fires. What if you gave all the firewood in replace for the water and you can still keep 1 food? | UV-Part, Coordination                          |
| <b>P1:</b> So I would get 3 water and 1 food and youd get 3 firewood and 2 food?                                                                                    | Non-strategic                                  |
| <b>P2:</b> Yea that seems like an alright trade to me 😊                                                                                                             | Non-strategic                                  |
| <b>P1:</b> Hmm... alright then 😊                                                                                                                                    | Non-strategic                                  |
| <b>P2:</b> Submit-Deal                                                                                                                                              |                                                |
| <b>P1:</b> Accept-Deal                                                                                                                                              |                                                |

Table 1: Sample dialogue from the CaSiNo dataset. **P1** and **P2** represent two participants in our study.

# Strategy Annotations

- Utterances are annotated with a specific negotiation strategy
- The annotated strategies correlate to the value of the negotiation
- Prosocial: Mindful of both parties
- Proself: Focus on individual performance
- A list of 9 strategies is defined, each falling into one of the falling categories:
  - Prosocial Generic
  - Prosocial About Preferences
  - Proself Generic
  - Proself About Preferences
  - Non-Strategic

| Label                              | Example                                                            | Count | $\alpha$ |
|------------------------------------|--------------------------------------------------------------------|-------|----------|
| <b>Prosocial Generic</b>           |                                                                    |       |          |
| Small-Talk                         | Hello, how are you today?                                          | 1054  | 0.81     |
| Empathy                            | Oh I wouldn't want for you to freeze                               | 254   | 0.42     |
| Coordination                       | Let's try to make a deal that benefits us both!                    | 579   | 0.42     |
| <b>Prosocial About Preferences</b> |                                                                    |       |          |
| No-Need                            | We have plenty of water to spare.                                  | 196   | 0.77     |
| Elicit-Pref                        | What supplies do you prefer to take the most of?                   | 377   | 0.77     |
| <b>Proself Generic</b>             |                                                                    |       |          |
| Undervalue-Partner                 | Do you have help carrying all that extra firewood? Could be heavy? | 131   | 0.72     |
| Vouch-Fairness                     | That would leave me with no water.                                 | 439   | 0.62     |
| <b>Proself About Preferences</b>   |                                                                    |       |          |
| Self-Need                          | I can't take cold and would badly need to have more firewood.      | 964   | 0.75     |
| Other-Need                         | we got kids on this trip, they need food too.                      | 409   | 0.89     |
| Non-strategic                      | Hello, I need supplies for the trip!                               | 1455  | -        |

Table 2: Utterance-level strategy annotations.  $\alpha$  refers to Krippendorff's alpha among 3 annotators on a subset of 10 dialogues ( $\sim 120$  utterances). An utterance can have multiple labels.

# Correlation Analysis

- Positive correlation between points scored and their own satisfaction ( $r=0.376$ ,  $p < 0.01$ )
- Positive correlation between points scored and their perception of the opponent ( $r=0.276$ ,  $p < 0.01$ )
- Positive correlation between joint points and the integrative potential of the utterance ( $r = 0.425$ ,  $p < 0.001$ )
- It is found that prosocial strategies relate to higher post-survey scores (satisfaction, and likeliness) and vice versa
- Proself strategies often lead to lower points scored
- Prosocial actions often lead to prosocial actions from the opponent

# Strategy Prediction

- Goal is to label Strategy types within utterances of negotiation
- BERT based encoder with a separately trained self-attention + fully connected layer for each strategy classification task (binary classification)
- Use sinusoidal positional embedding to represent turn index
- Fine tune each task-specific attention and FC block
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